PRELIMINARY WINE ANALYSIS

TO: E. and J. Gallo Winery
FROM: Barron Benbow
SUBJECT: CASE 1
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Introduction

Data and Analysis

This dataset was provided by a distributor as a source of potential wines for Gallo stores. There are 13,050 rows, each including information about the wine's name, country, reigion, winery, number ratings, year, type, price, and rating. We were to trying to find out what wine can Gallo sell at other shops. One key finding I found was that Red wine was cheaper than sparkling wine. I would suggest that Gallo should start selling Red Wine for the consistency and sparkling wines for the ratings and focusing on selling wine from the South American countries.

Section 1 – Discussion of Price and Rating

Wine Type	Count of Wines	Average Price	Average Rating
Red	8639	$38.66	3.89
Rosé	388	$12.64	3.74
Sparkling	272	$60.62	4.07
White	3751	$20.52	3.82
Overall	13050	33.13	3.87

Table 1: Wine type counts, average prices, and average ratings.
The largest difference in average price was between Sparkling and Rosé. The largest difference between the average ratings Rosé and Sparkling. The smallest difference between average rating was between red and white. The smallest difference in average price was between White Rosé .

For all wines, skewness of price was 13.85 indicating that the shape is right-tailed. There is a price extreme outlier, Pomerol 2012. Skewness of rating was -0.002 indicating that the shape was roughly symmetric. There are no extreme outliers for rating.

For all Red wine, skewness of price was 15.08 indicating that the shape is right-tailed. There is a price extreme outlier, Pétrus 2012. Skewness of rating was -0.06 indicating that the shape was roughly symmetric. There are no extreme outliers for rating.

For all Rosé, skewness of price was 9.55 indicating that the shape is right-tailed. There is a price extreme outlier, Gérad Bertrand 2018. Skewness of rating was .009 indicating that the shape was roughly symmetric. There are no extreme outliers for rating.

For all sparkling, skewness of price was 2.73 indicating that the shape is right-tailed. There is a price extreme outlier, Louis Roederer 2012. Skewness of rating was .11 indicating that the shape was roughly symmetric. There are no extreme outliers for rating.

For all white wine, skewness of price was 9.9 indicating that the shape is right-tailed. There is a price extreme outlier, Joseph Drouhin 2017. Skewness of rating was -0.04 indicating that the shape was roughly symmetric. There are no extreme outliers for rating.

Section 2 – Comparing South America vs Southern Hemisphere Average Wine Rating and Price

To compare prices between South America and southern hemisphere countries, the null hypothesis is that there is no difference between average wine prices in South American countries compared to southern hemisphere countries. The alternative hypothesis is that there is a difference in average wine prices in South American countries compared to southern hemisphere countries. The t-test statistic was t=-2.17 with a P-value of 0.030. The null hypothesis is rejected because the P-value is less than the 0.05 significance level. Based on the data, there is a statistically significant difference in average wine prices between South American countries and southern hemisphere countries.

Section 3 – Tableau Average Rating by Type and Map of Average Rating by Country

For Tableau Visualization, click here. There are two graphics - average rating by wine type, and average rating by country. On the graph of countries, you can use the filter to select countries to view. The country with the highest average rating was Moldova was 4.17, while the country with the lowest average rating was China at 3.20

Conclusion

Hypothesis test showed that there were was a significant difference in average price between South America and southern hemisphere countries. The hypothesis also shows that South Hemisphere has a signifcant difference in average rating compared to South American ratings. The Tableau visualization showed that Sparkling and Red Wine had the highest ratings. If you have any questions contact Barron Benbow at barronbenbow@arizona.edu

Appendix

Figure 1: Descriptive Statistics for all wines

Price		*Rating*	
Mean	33.12708812	Mean	3.86855172
Standard Error	0.615510888	Standard Error	0.00261511
Median	15.95	Median	3.9
Mode	9.9	Mode	3.8
Standard Deviation	70.31386905	Standard Deviation	0.29874102
Sample Variance	4944.040181	Sample Variance	0.0892462
Kurtosis	487.5179114	Kurtosis	0.09815069
Skewness	15.08484603	Skewness	-0.0021981
Range	3407.24	Range	2.4
Minimum	3.55	Minimum	2.5
Maximum	3410.79	Maximum	4.9
Sum	432308.5	Sum	50484.6
Count	13050	Count	13050
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Price		*Rating*	
Mean	38.65744994	Mean	3.889848362
Standard Error	0.882825035	Standard Error	0.003315572
Median	18.14	Median	3.9
Mode	9.9	Mode	3.8
Standard Deviation	82.05525078	Standard Deviation	0.308169934
Sample Variance	6733.06418	Sample Variance	0.094968708
Kurtosis	391.3635738	Kurtosis	0.003564765
Skewness	13.85057369	Skewness	-0.057134892
Range	3407.24	Range	2.3
Minimum	3.55	Minimum	2.5
Maximum	3410.79	Maximum	4.8
Sum	333961.71	Sum	33604.4
Count	8639	Count	8639
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Figure 2: Red Wine Descriptive Statistics

Price		*Rating*	
Mean	12.63987113	Mean	3.74226804
Standard Error	0.822192127	Standard Error	0.01394667
Median	8.945	Median	3.75
Mode	7.9	Mode	3.8
Standard Deviation	16.19530669	Standard Deviation	0.27471763
Sample Variance	262.2879589	Sample Variance	0.07546978
Kurtosis	123.1632356	Kurtosis	0.68938345
Skewness	9.550333254	Skewness	0.09184821
Range	245.3	Range	2.1
Minimum	3.7	Minimum	2.7
Maximum	249	Maximum	4.8
Sum	4904.27	Sum	1452
Count	388	Count	388
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Figure 3: Rosé Descriptive Statistics

Price		*Rating*	
Mean	60.61915441	Mean	4.065073529
Standard Error	4.520695372	Standard Error	0.017322309
Median	35.135	Median	4.1
Mode	14.9	Mode	3.9
Standard Deviation	74.55721808	Standard Deviation	0.285686837
Sample Variance	5558.778768	Sample Variance	0.081616969
Kurtosis	9.001643864	Kurtosis	-0.580743234
Skewness	2.727004978	Skewness	0.117996081
Range	488.5	Range	1.4
Minimum	6.5	Minimum	3.3
Maximum	495	Maximum	4.7
Sum	16488.41	Sum	1105.7
Count	272	Count	272
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Figure 4: Sparkling Wine Descriptive Statistics

Price		*Rating*	
Mean	20.51562517	Mean	3.818315116
Standard Error	0.500058587	Standard Error	0.004322647
Median	13.15	Median	3.8
Mode	9.9	Mode	3.8
Standard Deviation	30.62629221	Standard Deviation	0.264742265
Sample Variance	937.9697744	Sample Variance	0.070088467
Kurtosis	148.088236	Kurtosis	0.329565656
Skewness	9.923126131	Skewness	-0.049800827
Range	677.63	Range	2.2
Minimum	3.74	Minimum	2.7
Maximum	681.37	Maximum	4.9
Sum	76954.11	Sum	14322.5
Count	3751	Count	3751
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Figure 5: White Wine Descriptive Statistics

t-Test: Two-Sample Assuming Unequal Variances		
	S America Price	*S Hemishpere Price*
Mean	19.7352986	22.91655199
Variance	599.6050053	1814.947154
Observations	787	1308
Hypothesized Mean Difference	0	
df	2089	
t Stat	-2.16986871	
P(T<=t) one-tail	0.015064586	
t Critical one-tail	1.645583378	
P(T<=t) two-tail	0.030129172	
t Critical two-tail	1.961100233	
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Figure 6: Hypothesis test comparing prices

Figure 7: Hypothesis test comparing ratings

t-Test: Two-Sample Assuming Unequal Variances		
	S America Rating	*S Hemisphere Rating*
Mean	3.746378653	3.805122324
Variance	0.113762314	0.090876573
Observations	787	1308
Hypothesized Mean Difference	0	
df	1513	
t Stat	-4.01535922	
P(T<=t) one-tail	3.1132E-05	
t Critical one-tail	1.645861365	
P(T<=t) two-tail	6.2264E-05	
t Critical two-tail	1.961533144	
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